

Actuarial Pricing Analysis of Motor Insurance Claim Frequency: A Comparison of Generalized Linear Models and Machine Learning Approaches

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Abstract

Motor insurance pricing relies on accurate claim frequency estimation to support risk differentiation and premium adequacy. This study presents a comparative analysis of parametric and machine learning approaches using the French Motor Third-Party Liability (MTPL) dataset, comprising 678,013 policy records. A distributional assessment confirmed mild overdispersion (variance-to-mean ratio of 1.083), motivating the estimation of a Poisson GLM, a Negative Binomial GLM, a Random Forest, and an XGBoost model. Unlike an earlier working version of this analysis, the Negative Binomial dispersion parameter (α) is estimated directly from the data via maximum likelihood rather than fixed by assumption; α converged to 0.985 (SE = 0.049) on the primary split and remained stable (mean = 0.965, SD = 0.033 across 20 independent train-test splits). Bonus-Malus Score, Driver Age, Vehicle Age, and Population Density were statistically significant predictors ($p < 0.001$) in both GLMs, with Bonus-Malus Score the dominant predictor across all four models. To assess whether model rankings reflect genuine, reproducible differences rather than the outcome of a single train-test split, all four models were re-evaluated across 20 independent 80/20 splits. Random Forest achieved the lowest mean MAE (0.0982) and won 18 of 20 splits; XGBoost won the remaining 2; the Poisson and Negative Binomial GLMs never won a single split. The ranking was therefore highly reproducible, though the absolute performance gap between the best and weakest model was small (MAE range: 0.0982–0.0995). The Negative Binomial model achieved a lower, correctly-specified AIC than the Poisson model (228,839 vs. 229,549), confirming a superior likelihood-based fit consistent with the observed overdispersion; the Poisson model nonetheless achieved marginally better predictive accuracy by MAE, illustrating that AIC and MAE evaluate different objectives (statistical fit versus point-prediction error) and need not favour the same model. Overall, results indicate that machine learning methods offer a small, statistically reproducible predictive advantage over GLMs on this dataset, while GLMs retain their practical value in actuarial pricing through interpretability, statistical inference, and regulatory transparency. Model selection should therefore be guided by whether predictive accuracy or interpretability is the primary objective.

Keywords: Motor Insurance, Claim Frequency, Poisson GLM, Negative Binomial GLM, Random Forest, XGBoost, Bonus-Malus Score, Repeated-Split Validation, Actuarial Pricing, Model Comparison, French MTPL

1. Introduction

The accurate modelling of insurance claim frequency is a cornerstone of actuarial pricing practice. By estimating the expected number of claims a policyholder is likely to generate, insurers can stratify risk more

precisely, set premiums that reflect individual risk profiles, and maintain long-term portfolio profitability (Ohlsson & Johansson, 2010; Denuit et al., 2007). As motor insurance markets become increasingly data-driven, the range of modelling approaches available to actuaries has expanded considerably from traditional Generalized Linear Models (GLMs) to sophisticated machine learning algorithms (Henckaerts et al., 2021). A central question has emerged from this shift: do machine learning methods offer meaningful predictive gains over established parametric approaches, and if so, at what cost to interpretability and regulatory defensibility?

Studies consistently confirm that insurers across international motor insurance markets rely primarily on variables such as driver age, claim history, vehicle age, and geographic characteristics when pricing Motor Third-Party Liability (TPL) products (Antonio & Valdez, 2012; Denuit et al., 2007). These findings reflect established international practice and highlight the continued relevance of traditional rating factors in actuarial pricing. However, the statistical validation of these variables and a rigorous comparison of the modelling frameworks used to incorporate them remains an area of active research, particularly in the context of emerging insurance markets.

GLMs, and Poisson GLMs in particular, have long served as the actuarial standard for claim frequency modelling due to their suitability for count-based response variables, their interpretable coefficient structure, and their capacity to incorporate policy exposure as an offset term (Goldburd et al., 2025; Ohlsson & Johansson, 2010). The Negative Binomial GLM extends this framework by introducing a dispersion parameter to accommodate overdispersion, a common feature of insurance claim data where the variance exceeds the mean (Denuit et al., 2007). More recently, machine learning methods such as Random Forest and XGBoost have attracted attention in actuarial applications for their ability to capture complex, nonlinear relationships between predictors and outcomes without distributional assumptions (Kumar et al., 2024).

This study addresses three interrelated objectives. First, it develops and validates a claim frequency model using the French Motor Third-Party Liability (MTPL) dataset, a well-established benchmark for actuarial research (Noll et al., 2018). Four models are estimated and compared: a Poisson GLM, a Negative Binomial GLM with a maximum-likelihood-estimated dispersion parameter, a Random Forest, and an XGBoost regressor. Second, the study examines the consistency of variable importance findings across these models as a means of cross-validating the relevance of the selected rating variables. Third, recognising that a single train-test split cannot, on its own, establish whether an observed model ranking is reproducible, the study repeats model estimation and evaluation across 20 independent random splits to quantify ranking stability directly — a validation step motivated by long-standing evidence that single-split evaluation can be an unreliable guide to model selection (Kohavi, 1995).

The explanatory variables examined are Bonus-Malus Score, Driver Age, Vehicle Age, and Population Density, factors identified as statistically relevant in the French MTPL dataset (Noll et al., 2018) and widely documented as key pricing factors in motor insurance markets internationally (Antonio & Valdez, 2012). A distributional assessment of the response variable was conducted prior to model estimation to evaluate the appropriateness of the Poisson assumption and justify the inclusion of the Negative Binomial specification.

The findings contribute to the growing literature on the comparative performance of statistical and machine learning methods in actuarial pricing and offer practical insights for insurers seeking to balance predictive accuracy with the interpretability requirements of actuarial practice and insurance regulation. The remainder of this paper is structured as follows: Section 2 reviews the relevant literature; Section 3 describes the dataset and variable selection; Section 4 outlines the methodology, including the repeated-split validation procedure; Section 5 presents the results; and Section 6 discusses the findings, limitations, and directions for future research.

2. Literature Review

2.1 Claim Frequency Modelling in Motor Insurance

Claim frequency modelling is a key component of actuarial pricing because it allows insurers to estimate the expected number of future claims and determine premiums that reflect the underlying risk of policyholders (Goldburd et al., 2025; Ohlsson & Johansson, 2010). Modern pricing models typically use policyholder, vehicle, and geographic characteristics to explain variations in claim occurrence. The Poisson distribution has historically been the default choice for modelling claim counts given its suitability for non-negative integer-valued data, though the assumption of equidispersion is frequently violated in practice (Tzougas, 2020). According to the Saudi Insurance Authority (2024), insurers increasingly rely on data-driven pricing approaches and risk-based rating variables to improve pricing accuracy and portfolio profitability. Wuthrich & Merz (2023) provide benchmark results on the French MTPL dataset against which machine learning approaches are routinely evaluated in the actuarial literature.

2.2 Generalized Linear Models in Actuarial Pricing

Generalized Linear Models (GLMs) are among the most widely used actuarial techniques for modelling insurance claims because they provide an interpretable relationship between claim outcomes and risk factors (de Jong & Heller, 2008). For claim frequency applications, Poisson GLMs are commonly used since the response variable represents count data. Where claim data exhibit overdispersion, that is, where the variance exceeds the mean, the Negative Binomial GLM provides a more flexible alternative by introducing a dispersion parameter to accommodate this excess variability (Ismail & Jemain, 2007). Goldburd et al. (2025) highlight that exposure can be incorporated as an offset term, allowing fair comparison between policyholders with different levels of risk exposure.

Using the French Motor Third-Party Liability (MTPL) dataset, Noll et al. (2018) applied a Poisson-based modelling framework to predict claim frequency using variables such as driver age, vehicle age, Bonus-Malus score, population density, and regional characteristics. Their work demonstrates the practical application of GLMs in motor insurance pricing and provides a framework similar to that adopted in this study. Wuthrich & Merz (2023) further develop GLM benchmarks on the same dataset, providing a widely used reference point for actuarial machine learning comparisons.

2.3 Rating Variables in Motor Insurance Pricing

The Saudi Insurance Authority (2024) identified several commonly used rating variables in Motor Third-Party Liability (TPL) pricing across Saudi insurers (Figure 1). Driver age, vehicle age, geographical region, and claim-history-related measures are among the most frequently used pricing factors. These variables closely align with the predictors available in the French MTPL dataset, particularly Driver Age, Vehicle Age, Bonus-Malus Score, and Population Density. The consistency between industry practice and the variables selected for this study supports their relevance for claim frequency modelling.

Driver age is among the most widely documented rating variables in motor insurance, with younger drivers consistently associated with higher claim frequencies across multiple markets and datasets (Dionne & Vanasse, 1989). Vehicle age has been found to have an inverse relationship with claim frequency in several studies, potentially reflecting differences in vehicle value, usage patterns, or policyholder behaviour (Ohlsson & Johansson, 2010). Bonus-Malus Score serves as the primary a posteriori rating factor in many European motor insurance markets and has been shown to be among the strongest predictors of future claim frequency (Denuit et al., 2007; Noll et al., 2018). Population density captures geographic exposure, with urban policyholders generally exhibiting higher claim frequencies due to increased traffic congestion and vehicle interaction (Goldburd et al., 2025).

2.4 Machine Learning Methods in Insurance Pricing

Tree-based ensemble methods, particularly Random Forest (Breiman, 2001) and XGBoost (Chen & Guestrin, 2016), have demonstrated strong predictive performance on structured insurance data. Henckaerts et al. (2021) developed full tariff plans using tree-based machine learning for both frequency and severity of claims, demonstrating that ensemble models can achieve comparable accuracy to GLMs while providing complementary variable importance measures. Their study, conducted on the French MTPL dataset, found that experience-rating variables consistently ranked highest in both Random Forest and gradient boosting models, a finding directly relevant to the results of this study.

GLMs often remain competitive with machine learning when datasets exhibit limited nonlinear structure, and Kumar et al. (2024) similarly found that performance advantages of ML methods diminish for structured tabular insurance data with few predictors. While existing studies have applied GLMs to the French MTPL dataset and demonstrated their effectiveness for claim frequency modelling, few have conducted a systematic comparison of parametric GLMs and machine learning models within the same analytical framework using identical data splits, evaluation metrics, and — critically — a direct test of whether reported performance differences are reproducible across resamples of the data rather than an artefact of a single split. Furthermore, the relevance of these modelling approaches to Gulf insurance markets, and the Saudi market in particular, has received limited attention in the academic literature. This study addresses these gaps by estimating and comparing four models on a common, repeatedly-validated evaluation framework and situating the findings within the Saudi motor insurance pricing context as documented by the Insurance Authority (2024).

2.5 Model Validation and Interpretability

A model comparison based on a single train-test split cannot, by itself, distinguish a genuine performance difference from sampling variation attributable to that particular split. Kohavi (1995) provides the classical demonstration of this issue, comparing cross-validation and bootstrap estimators of accuracy and showing that repeated resampling is necessary to obtain a reliable estimate of a model's true generalisation performance and of the uncertainty around it. This motivates the repeated-split validation procedure adopted in Section 4.8 of this study. Separately, as machine learning methods have grown more prevalent in insurance pricing, the tension between predictive accuracy and interpretability has become a central practical concern, since actuarial pricing decisions must typically be justified to regulators and other stakeholders (Goldburd et al., 2025). Model-agnostic interpretability tools such as SHAP (SHapley Additive exPlanations) have been developed to help bridge this gap by attributing individual predictions to specific input features in a theoretically grounded way (Lundberg & Lee, 2017), and are noted in Section 6 as a natural extension of the present work.

3. Data Description

3.1 Data Source

The dataset used in this study is the French Motor Third-Party Liability (MTPL) insurance dataset (freMTPL2freq), a publicly available benchmark dataset widely used in actuarial research (Noll et al., 2018; Henckaerts et al., 2021). It contains policy-level information on motor insurance contracts and claim experience, including policyholder characteristics, vehicle attributes, exposure, and claim counts. The dataset was accessed from Kaggle (<https://www.kaggle.com/datasets/floser/french-motor-claims-datasets-fremtpl2freq>) and comprises 678,013 policy records.

3.2 Selected Variables

Several variables are available within the dataset; however, this study focuses on a subset relevant to motor insurance pricing and risk assessment. The response variable is ClaimNb, representing the number of claims reported by each policyholder during the exposure period. Exposure records the fraction of the year during which the policy was active and is incorporated as an offset term in the GLMs and as a standard input feature in the machine learning models. The four explanatory variables are: Driver Age (DrivAge), the age of the insured driver in years; Vehicle Age (VehAge), the age of the insured vehicle in years; Bonus-Malus Score (BonusMalus), an experience-rating index reflecting accumulated claims history, where higher scores indicate a worse claims record; and Population Density (Density), the population density of the policyholder's residential area in inhabitants per square kilometre. These variables were selected based on their theoretical relevance to motor insurance risk and their established use in prior actuarial studies on the French MTPL dataset (Noll et al., 2018).

3.3 Data Preparation

Prior to analysis, the dataset was reviewed for completeness and consistency. No missing values were identified across the selected variables, and no non-positive Exposure values or duplicate policy records were found. For exploratory data analysis (EDA), continuous variables were grouped into meaningful intervals to facilitate visualization and interpretation of claim frequency patterns across different risk segments. Risk classification is intended to group individual risks with reasonably similar expectations of loss, forming the actuarial foundation for the rating variable selection approach adopted in this study (American Academy of Actuaries, 2025). Driver Age, Vehicle Age, Bonus-Malus Score, and Population Density were each categorised into ranges aligned with common actuarial practice. These grouped variables were used solely for descriptive and graphical purposes; all four models were estimated using the original, ungrouped variable values to preserve the full information content of the data.

3.4 Summary Statistics

Table 1. Summary Statistics of Variables Used in the Analysis

Variable	Mean	Std. Dev.	Min	Median	Max
Claim Frequency	0.053	0.240	0.000	0.000	16.000
Exposure	0.530	0.360	0.003	0.490	2.010
Driver Age	45.500	14.140	18.000	44.000	100.000
Vehicle Age	7.040	5.670	0.000	6.000	100.000
Bonus-Malus Score	59.760	15.640	50.000	50.000	230.000
Population Density	1,792.42	3,958.65	1.000	393.000	27,000.000

The dataset contains 678,013 policy records. The average claim frequency is 0.053 claims per policy, while the median value is zero, indicating that most policyholders did not report a claim during the observation period. Exposure ranges from approximately 0.003 to 2.01 policy years, reflecting differences in coverage duration across policies; no policy has non-positive exposure. The average driver age is 45.5 years, while the average vehicle age is 7.0 years. Bonus-Malus Scores range from 50 to 230, indicating substantial variation in policyholder claim histories. Population density exhibits considerable variation, ranging from 1 to 27,000, suggesting representation from both rural and highly urbanised areas. The maximum claim frequency of 16.00 reflects a small number of high-frequency policies, characteristic of heavy-tailed insurance claim distributions (Denuit et al., 2007). Overall, the dataset contains substantial variability across the explanatory variables, making it suitable for investigating factors associated with motor insurance claim frequency.

4. Methodology

4.1 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the dataset and examine the relationships between claim frequency and the selected explanatory variables. For each variable, claim frequency was calculated and visualised across grouped intervals to identify potential trends and patterns. The purpose of this step was to provide an initial understanding of the data and support the selection of variables for subsequent statistical modelling, following standard actuarial practice (Goldburd et al., 2025).

4.2 Correlation Analysis

Following the exploratory analysis, a correlation analysis was performed to examine the strength and direction of relationships among the numerical variables. Pearson correlation coefficients were calculated and summarised using a correlation matrix. The objective was to identify potential linear relationships between explanatory variables and to assess the presence of multicollinearity prior to model development, as highly correlated predictors can affect the stability and interpretation of regression models (Goldburd et al., 2025).

4.3 Distributional Assessment of the Response Variable

Prior to specifying the Poisson GLM, a distributional assessment of ClaimNb was conducted to evaluate whether the Poisson assumption of equidispersion holds. The variance-to-mean ratio (dispersion index) was computed across the full dataset. A ratio exceeding 1.0 indicates overdispersion, which if unaddressed may lead to underestimated standard errors and invalid inference (Denuit et al., 2007; Goldburd et al., 2025). The proportion of zero observations was also reported, as high zero-inflation is characteristic of MTPL datasets. Notably, a Poisson distribution with the sample mean claim frequency (0.0532) alone predicts a zero-claim proportion of approximately 94.8%, closely matching the observed 94.97% — indicating that the high zero-count is largely explained by the low mean rate itself rather than by additional zero-inflation beyond what a (over-dispersed) Poisson process already implies.

4.4 Poisson Generalized Linear Model

To model claim frequency, a Poisson Generalized Linear Model (GLM) was developed. Claim frequency represents count data, making the Poisson distribution a natural starting point for modelling the number of claims reported by each policyholder. Let N_i denote the number of claims reported by policyholder i . The response variable is assumed to follow a Poisson distribution, $N_i \sim \text{Poisson}(\mu_i)$, with a log-link function relating the expected claim frequency to the explanatory variables (Nelder & Wedderburn, 1972):

$$\log(\mu_i) = \beta_0 + \beta_1(\text{BonusMalus}_i) + \beta_2(\text{DrivAge}_i) + \beta_3(\text{VehAge}_i) + \beta_4(\text{Density}_i) + \log(\text{Exposure}_i)$$

where $\log(\text{Exposure}_i)$ is an offset term adjusting for differences in policy duration. Model parameters were estimated using Maximum Likelihood Estimation (MLE).

4.5 Negative Binomial GLM with Estimated Dispersion Parameter

While the Poisson GLM is widely used for modelling count data, it assumes that the mean and variance of the response variable are equal. Section 5.3.1 confirms mild overdispersion in this dataset (variance-to-mean ratio of 1.083); ignoring overdispersion may lead to underestimated standard errors and overly optimistic statistical inference (Ismail & Jemain, 2007). A Negative Binomial GLM (NB2 parameterisation, where the variance is $\mu_i + \alpha\mu_i^2$) was therefore also developed, retaining the same log-link specification, explanatory variables, and exposure offset as the Poisson model:

$$N_i \sim \text{NegativeBinomial}(\mu_i, \alpha)$$

The dispersion parameter $\alpha \geq 0$ is estimated directly from the training data by maximum likelihood, rather than fixed by assumption. Because a joint optimisation over the regression coefficients and α proved numerically

unstable for this dataset under standard solvers, α was instead estimated using an alternating profile-likelihood procedure — the same approach underlying the widely used `glm.nb` implementation in R (Venables & Ripley, as implemented in the MASS package): the model alternates between (i) a one-dimensional maximum-likelihood search for α given the current fitted values, and (ii) refitting the GLM regression coefficients by iteratively reweighted least squares at the updated α , until α stabilises. This procedure is a standard, numerically stable alternative to joint estimation and yields a genuine maximum-likelihood estimate of α , together with its standard error obtained from the curvature of the profile log-likelihood at convergence.

On the primary 80/20 split, this procedure converged to $\alpha = 0.985$ (SE = 0.049; approximate 95% CI [0.889, 1.081]). To confirm this estimate was not an artefact of one particular split, the same procedure was re-applied across the 20 independent splits described in Section 4.8; α converged in every split (mean = 0.965, SD = 0.033, range [0.880, 1.011]), indicating a stable, well-identified dispersion parameter rather than a numerically fragile one. Because α is now estimated from the data, it is treated as an additional free parameter (contributing one degree of freedom) when computing the model's AIC, as detailed in Section 5.3.5.

4.6 Machine Learning Models

To complement the parametric GLM framework, two machine learning models were estimated: a Random Forest Regressor (Breiman, 2001) and an XGBoost Regressor (Chen & Guestrin, 2016). Both models were applied to the same four explanatory variables used in the GLMs, with Exposure included as a standard input feature rather than an offset term, consistent with the non-parametric nature of tree-based methods (Henckaerts et al., 2021).

The Random Forest was configured with 100 estimators, a maximum depth of 10, and a fixed random state (`random_state = 42`) to ensure reproducibility. The XGBoost model was estimated with 300 estimators, a learning rate of 0.05, maximum depth of 4, and subsampling rates of 0.8. Both models were trained on the identical 80/20 train-test split applied to the GLMs. Feature importance scores were extracted from both models to enable a model-agnostic assessment of variable relevance.

4.7 Primary Train-Test Validation

To evaluate predictive performance, the dataset was randomly divided into training and testing subsets using an 80/20 split (542,410 training observations; 135,603 testing observations), with `random_state = 42` ensuring identical splits across all four models. Predictive performance was assessed using Mean Absolute Error (MAE) and mean Poisson deviance. For the GLMs, model fit was additionally assessed using the Akaike Information Criterion (AIC), where lower values indicate a better balance between model fit and complexity. The use of a held-out test set provides an objective measure of out-of-sample predictive performance and reduces the risk of overfitting (Goldburd et al., 2025).

4.8 Repeated-Split Validation

A single train-test split, however carefully constructed, cannot by itself establish whether an observed model ranking reflects a genuine, reproducible difference in performance or the outcome of a favourable (or unfavourable) split for a particular model (Kohavi, 1995). To address this, all four models were re-estimated and re-evaluated across 20 independent 80/20 splits of the full dataset, using fixed, reproducible random seeds (0 through 19) for the split itself while retaining identical predictors, formulas, exposure treatment, and hyperparameters to the primary analysis in every split. For each split and model, test MAE, mean Poisson deviance, convergence status, and, for the Negative Binomial model, the estimated α , were recorded. Exposure was confirmed strictly positive in both the training and test fold of every split prior to fitting. From these 20 runs, the mean, standard deviation, and 95% confidence interval of MAE were computed for each

model, together with the number of splits "won" (lowest MAE) by each model and each model's average rank across splits. This procedure — sometimes referred to as repeated random sub-sampling or Monte Carlo cross-validation — directly answers the question of whether a reported performance ranking is reproducible, complementing rather than replacing the primary single-split analysis reported in Section 4.7.

5. Results

5.1 Exploratory Data Analysis Results

(Results unchanged from the primary analysis; see Figures 2–5.)

5.1.1 Driver Age

The highest claim frequency was observed among drivers aged 18-24, while lower frequencies were observed among drivers aged 25-44. Claim frequency increases slightly for older age groups but remains below the level observed for the youngest drivers, consistent with the elevated risk associated with inexperience widely reported in actuarial pricing research (Antonio & Valdez, 2012).

5.1.2 Vehicle Age

Claim frequency is highest for vehicles aged 0-2 years and declines monotonically with increasing vehicle age, potentially reflecting differences in vehicle value, usage intensity, or behavioural differences among owners of newer versus older vehicles.

5.1.3 Bonus-Malus Score

Policyholders classified as high risk exhibit the highest claim frequency, followed by those in the medium risk category, while the low-risk group has the lowest claim frequency — a monotonic pattern consistent with the actuarial rationale for experience rating (Denuit et al., 2007).

5.1.4 Population Density

Policyholders residing in high-density areas exhibit the highest claim frequency, followed by medium- and low-density areas respectively, consistent with greater traffic congestion and vehicle interaction in urban environments (Su & Bai, 2020).

5.2 Correlation Analysis Results

Correlations between claim frequency and the explanatory variables are relatively weak, with no single variable exhibiting a strong linear relationship with the response. Bonus-Malus Score shows the strongest positive correlation with claim frequency (0.051), followed by Population Density (0.011) and Driver Age (0.011). The strongest correlation among the explanatory variables is between Driver Age and Bonus-Malus Score (−0.48): older drivers tend to have lower (better) Bonus-Malus scores, consistent with the accumulation of no-claims discounts over a longer driving history under a typical bonus-malus system, while younger, less experienced drivers have had less time to earn such a discount. This moderate negative relationship remains below commonly used thresholds for multicollinearity concern, suggesting that the selected variables can be jointly included in the GLM without material stability issues (Goldburd et al., 2025).

5.3 Statistical Model Results

5.3.1 Distributional Assessment

Table 2. Distributional Statistics for ClaimNb

Statistic	Value
Mean (ClaimNb)	0.05324

Statistic	Value
Variance (ClaimNb)	0.05765
Variance-to-Mean Ratio (Dispersion Index)	1.083
Proportion of Zero Claims	94.97%

The variance-to-mean ratio of 1.083 indicates mild overdispersion; the variance exceeds the mean by approximately 8.3%, representing a slight but detectable departure from the Poisson equidispersion assumption. These findings motivated the estimation of both a Poisson GLM and a Negative Binomial GLM with an estimated dispersion parameter (Section 4.5).

5.3.2 Poisson GLM Results

Table 3. Poisson GLM Coefficient Estimates

Variable	Coefficient	p-value
Intercept	-3.7115	<0.001
Bonus-Malus Score	0.0229	<0.001
Driver Age	0.0069	<0.001
Vehicle Age	-0.0428	<0.001
Population Density	9.683×10^{-5}	<0.001

All explanatory variables are statistically significant at the 1% level. The coefficient associated with Bonus-Malus Score is positive, indicating that policyholders with higher scores tend to exhibit higher expected claim frequencies. Vehicle Age exhibits a negative coefficient, implying that older vehicles are associated with lower expected claim frequencies. Driver Age and Population Density both carry positive coefficients.

5.3.3 Poisson Model Performance

The fitted model achieved a Mean Absolute Error (MAE) of 0.0989 and a mean Poisson deviance of 0.3217 on the held-out test set (135,603 observations). The model also produced a Cox-Snell Pseudo R^2 of approximately 0.0099. Although low, this is expected for claim frequency modelling, since claim occurrences are rare events influenced by many unobserved factors; predictive performance measures and coefficient significance are therefore generally more informative than R^2 for evaluating Poisson GLMs in this setting (Goldburd et al., 2025).

5.3.4 Negative Binomial GLM Results (α Estimated)

Table 4. Negative Binomial GLM (NB2) Coefficient Estimates — α Estimated via Profile-Likelihood MLE

Variable	Coefficient	p-value
Intercept	-3.7398	<0.001
Bonus-Malus Score	0.0233	<0.001
Driver Age	0.0073	<0.001
Vehicle Age	-0.0444	<0.001
Population Density	1.045×10^{-5}	<0.001
Estimated α	0.9846 (SE = 0.0490)	—

All explanatory variables remain statistically significant at the 1% level, with coefficient directions and magnitudes essentially unchanged from the Poisson specification (the largest difference is 0.0004 for Driver Age), confirming that the mild overdispersion identified in Section 5.3.1 does not materially distort the point estimates. The estimated dispersion parameter, $\alpha = 0.9846$ (95% CI [0.889, 1.081]), is a genuine

data-derived estimate rather than an assumed value, and — as reported in Section 4.5 — remained stable when the estimation procedure was repeated across 20 independent data splits (mean α = 0.965, SD = 0.033, all 20 splits converged).

5.3.5 Negative Binomial Model Performance and the AIC Correction

Because α is now estimated from the data rather than fixed, it is counted as an additional estimated parameter when computing AIC (six parameters in total: the intercept, four coefficients, and α), rather than the five parameters used when α is treated as given. Using the joint log-likelihood evaluated at the estimated coefficients and α , the Negative Binomial model achieves an AIC of **228,839.46**, compared with **229,548.83** for the Poisson model — a difference of approximately 709 points that confirms a statistically superior likelihood-based fit, consistent with the mild overdispersion documented in Section 5.3.1.

On predictive accuracy, however, the Negative Binomial model achieved a test MAE of **0.0994** and mean Poisson deviance of 0.3220, both marginally worse than the Poisson model's MAE of 0.0989 and deviance of 0.3217. This pattern — the Negative Binomial model winning on AIC while the Poisson model achieves marginally better out-of-sample MAE — was confirmed to be consistent, not a one-off result: across all 20 splits of the repeated-split validation described in Section 5.5, the Negative Binomial model never achieved the lowest test MAE among the four models. This is not a contradiction: AIC evaluates in-sample likelihood-based statistical fit, appropriately rewarding the Negative Binomial model for correctly modelling the extra dispersion in the data, whereas MAE evaluates out-of-sample point-prediction accuracy, a different and equally valid but distinct objective. **The Negative Binomial model should therefore not be interpreted as the best-performing model for point prediction; its value lies in providing a more statistically defensible characterisation of the claim-count distribution, including more appropriate standard errors and inference, which the Poisson model understates under overdispersion.**

5.4 Machine Learning Model Results

5.4.1 Predictive Performance (Primary Split)

Table 5. Predictive Performance and Model Selection Criteria (Primary 80/20 Split)

Model	MAE	AIC	Model Type
Poisson GLM	0.0989	229,548.83	Parametric
Negative Binomial GLM (α estimated)	0.0994	228,839.46	Parametric
Random Forest	0.0981	—	Non-parametric
XGBoost	0.0982	—	Non-parametric

Note: AIC is not applicable to machine learning models.

On this primary split, Random Forest achieved the lowest MAE (0.0981) and XGBoost the second lowest (0.0982), both modestly below the Poisson GLM (0.0989) and Negative Binomial GLM (0.0994). The total spread across all four models is 0.0013 MAE units. Section 5.5 reports whether this ordering is reproducible across independent splits.

5.4.2 Feature Importance Analysis

Table 6. Random Forest Feature Importance Rankings

Rank	Feature	Importance Score
1	Bonus-Malus Score	0.2655
2	Exposure	0.2340

Rank	Feature	Importance Score
3	Driver Age	0.1719
4	Population Density	0.1714
5	Vehicle Age	0.1573

Table 7. XGBoost Feature Importance Rankings

Rank	Feature	Importance Score
1	Bonus-Malus Score	0.3164
2	Exposure	0.2683
3	Vehicle Age	0.2286
4	Driver Age	0.1209
5	Population Density	0.0658

Bonus-Malus Score consistently ranked as the most important predictor across both machine learning models, corroborating the GLM coefficient estimates where Bonus-Malus Score exhibited the largest positive standardised effect. Exposure ranked second in both ML models, reflecting the critical role of policy duration in determining claim counts. The convergence of variable rankings across three independent modelling frameworks (Poisson/NB2, Random Forest, XGBoost) provides strong cross-validation evidence for the relevance of the selected rating variables (Henckaerts et al., 2021).

5.5 Repeated-Split Validation Results

To determine whether the ranking observed on the primary split (Section 5.4.1) reflects a genuine, reproducible performance difference, all four models were re-estimated and re-evaluated across 20 independent 80/20 splits (Section 4.8). All 80 model-split combinations (4 models × 20 splits) completed successfully with no convergence failures.

Table 8. Repeated-Split Validation Summary (n = 20 independent splits)

Model	Mean MAE	Std. Dev. (MAE)	Wins (out of 20)	Win %	Average Rank
Poisson GLM	0.09897	0.00040	0	0%	3.0
Negative Binomial GLM (α estimated)	0.09952	0.00039	0	0%	4.0
Random Forest	0.09820	0.00040	18	90%	1.1
XGBoost	0.09828	0.00039	2	10%	1.9

Rank 1 = lowest MAE (best) in a given split; Average Rank is the mean of a model's per-split rank across all 20 splits. Detailed per-split results, including 95% confidence intervals on mean MAE, are provided in Appendix B.

Two findings emerge clearly from this analysis. First, the ranking is **highly reproducible**: Random Forest achieved the lowest MAE in 18 of 20 splits and XGBoost in the remaining 2, while the Poisson and Negative Binomial GLMs never achieved the lowest MAE in any of the 20 splits — the Poisson GLM ranked third and the Negative Binomial GLM ranked fourth in every single split, without exception. The 95% confidence intervals for mean MAE (Appendix B) do not overlap between the tree-based models and the GLMs, indicating this is a statistically distinguishable difference rather than sampling noise from a single, possibly favourable, split. Second, despite this reproducibility, the **absolute magnitude of the difference is small**: mean MAE ranges from 0.09820 (Random Forest) to 0.09952 (Negative Binomial GLM), a spread of roughly 0.0013, and the standard deviation of MAE within each model across splits (~0.0004) is of a similar order of magnitude to

the gap between the two tiers of models. In practical underwriting terms, this represents a modest, not transformative, predictive advantage for the machine learning models.

Taken together, these results support two conclusions that are not in tension with one another: the machine learning models provide a real, reproducible predictive edge over both GLM specifications on this dataset, and that edge is numerically small enough that the interpretability, statistical inference, and regulatory transparency advantages of the GLMs (Section 6) remain highly relevant considerations in model selection.

5.6 Model Selection Considerations

Given the modest but reproducible predictive gap documented above, the selection of an appropriate model for actuarial pricing purposes should be guided by criteria beyond MAE alone. The Poisson GLM and Negative Binomial GLM offer three advantages particularly relevant in an insurance pricing context. First, they produce interpretable coefficient estimates that quantify the directional and magnitude effects of each rating variable on expected claim frequency. Second, parametric models provide statistical inference tools such as standard errors, p-values, and confidence intervals — including, now, a properly estimated dispersion parameter for the Negative Binomial specification — which support regulatory scrutiny and actuarial justification. Third, the Poisson GLM explicitly accounts for policy exposure through an offset term, ensuring that model outputs represent claim rates rather than raw counts. Between the two GLMs, the Negative Binomial specification is preferred where a statistically correct treatment of overdispersion is the priority (e.g., for inference and standard errors), while the Poisson specification achieved marginally better point-prediction accuracy in this study; in practice the two are close enough (Section 5.3.5) that either is defensible, and the choice should be stated explicitly along with its rationale. Consistent with the recommendation of Kumar et al. (2024), the Poisson GLM is identified as the preferred default model for actuarial pricing applications in this study on the basis of predictive accuracy and simplicity, with the Negative Binomial specification recommended as a robustness check wherever overdispersion is a material concern.

6. Discussion

6.1 Comparison with Literature

Table 9. Comparison of Key Findings with Published Literature

Variable / Finding	This Study	Noll et al. (2018)	Henckaerts et al. (2021)	Kumar et al. (2024)
Bonus-Malus / Claim History	Dominant predictor (all models)	Strong predictor	Highest-ranked feature	High importance
Driver Age	Significant (+)	Included	Included	Used in models
Vehicle Age	Significant (–)	Included	Included	Used in models
Population Density	Significant (+)	Included	Geographic feature	Used in models
GLM vs ML (MAE, single split)	ML slightly ahead (0.0981–0.0994)	GLM only	GLM competitive with ML	Performance gap narrow
GLM vs ML (repeated-split ranking)	ML ranking reproducible in 20/20 splits; gap numerically small	Not assessed	Not assessed	Not assessed
NB vs Poisson (AIC vs MAE)	NB wins AIC ($\Delta\text{AIC} \approx 709$); Poisson wins MAE	Poisson only	Poisson baseline	Not assessed

Bonus-Malus Score was identified as the dominant predictor of claim frequency in this study, consistent with Noll et al. (2018) and Henckaerts et al. (2021), who reported that claim-history and experience-rating variables consistently rank highest across GLM and tree-based frameworks in the French MTPL dataset. At a theoretical level, the primacy of Bonus-Malus Score is supported by its actuarial design as a posteriori pricing

factor that captures the portion of individual risk not explained by a priori rating variables (Antonio & Valdez, 2012; Denuit et al., 2007). The convergence across three modelling frameworks in this study strengthens this finding considerably.

Regarding the comparative performance of modelling approaches, this study extends Kumar et al. (2024) and Henckaerts et al. (2021) by directly testing, rather than assuming, whether a small observed performance gap between GLMs and machine learning models is reproducible. The repeated-split analysis (Section 5.5) confirms that it is: the machine learning models achieved the best predictive performance, consistently and reproducibly, but the margin of improvement over the GLMs was small in absolute terms. This nuance — a real but modest advantage — is a more precise characterisation than either "GLMs are equivalent to ML" or "ML clearly outperforms GLMs," and is only possible to state with confidence because of the repeated-split evidence.

6.2 Interpretation of Model Findings

The Poisson GLM coefficient estimates provide direct, quantifiable insights into the relationship between each rating variable and expected claim frequency. Each unit increase in Bonus-Malus Score is associated with a multiplicative increase in expected claims of approximately $e^{0.0229} \approx 1.023$, a 2.3% increase in expected claims per unit; given that Bonus-Malus Scores in this dataset range from 50 to 230, a policyholder at the upper end of the scale would be expected to generate substantially more claims than one at the minimum score, reinforcing the actuarial rationale for experience-based pricing. Vehicle Age carries a coefficient of -0.0428 (Poisson) / -0.0444 (Negative Binomial), corresponding to an approximate 4.2–4.4% reduction in expected claim frequency per additional year of vehicle age.

The Negative Binomial model's estimated dispersion parameter, $\alpha = 0.985$, is close to — but should not be conflated with — the value of 1 that an earlier, non-estimated version of this analysis had assumed by default; the estimation procedure in Section 4.5 confirms this is a genuine, well-identified, data-supported value rather than an arbitrary one, with a 95% confidence interval of $[0.889, 1.081]$ that is stable across repeated resampling of the data. An important and easily overlooked distinction demonstrated by this study is that a model can achieve a superior likelihood-based fit (as the Negative Binomial model does, by AIC) while a simpler model achieves marginally better out-of-sample point-prediction accuracy (as the Poisson model does, by MAE). Actuaries should therefore be explicit about which objective — statistical fit and correct inference, or point-prediction accuracy — is being optimised when comparing GLM specifications, rather than treating a single metric as universally decisive.

6.3 Limitations

Several limitations should be considered when interpreting these findings. First, the analysis is based exclusively on the French MTPL dataset, which reflects driving behaviour, traffic conditions, and regulatory structures specific to France; the transferability of quantitative findings to other markets, including Saudi Arabia, cannot be assumed without further validation using local data. Second, the study addresses only claim frequency; a complete actuarial pricing model requires a frequency-severity framework combining claim counts with claim costs to derive expected pure premiums. Third, while the repeated-split validation in Section 5.5 substantially strengthens confidence in the reported model ranking, all 20 splits are drawn from the same underlying dataset and time period; it does not address out-of-time or geographic generalisation, which the dataset's structure does not support testing directly. Fourth, the machine learning models were estimated with moderate, hand-selected hyperparameters rather than an exhaustive tuning search; more extensive optimisation could yield modest further improvements, though the consistency of the ranking across 20 splits suggests the qualitative conclusion is unlikely to be sensitive to further tuning. Fifth, the dataset does not include variables used in the Saudi context such as driver nationality, vehicle make and model, or marital status (Saudi Insurance Authority, 2024), which could improve model performance and better reflect the local

pricing environment.

7. Conclusion

This study estimated and compared four models for motor insurance claim frequency using the French MTPL dataset: a Poisson GLM, a Negative Binomial GLM with a maximum-likelihood-estimated dispersion parameter, a Random Forest, and an XGBoost regressor. All four explanatory variables were statistically significant ($p < 0.001$) in both GLMs, with Bonus-Malus Score emerging as the dominant predictor across all four modelling frameworks — independently confirmed by the Random Forest and XGBoost feature importance rankings and consistent with its actuarial role as the primary a posteriori rating factor (Denuit et al., 2007; Antonio & Valdez, 2012).

The Negative Binomial dispersion parameter was estimated directly from the data via a profile-likelihood maximum-likelihood procedure ($\alpha = 0.985$, stable across 20 independent data splits), replacing an earlier, non-estimated specification. The Negative Binomial model achieved a superior, correctly-specified AIC relative to the Poisson model (228,839 vs. 229,549), confirming a better likelihood-based fit consistent with the mild overdispersion documented in the data; the Poisson model, however, achieved marginally better predictive accuracy by MAE, a pattern confirmed to be consistent across repeated resampling rather than a feature of one split. These two results are complementary, not contradictory: they reflect the fact that AIC and MAE evaluate different objectives.

To establish whether the modest MAE advantage of the machine learning models over the GLMs on a single split was reproducible, all four models were re-evaluated across 20 independent train-test splits. The resulting ranking was highly stable — Random Forest and XGBoost won 18 and 2 of 20 splits respectively, while neither GLM won a single split — confirming that machine learning methods offer a real, reproducible predictive advantage on this dataset. That advantage, however, is numerically small (a MAE spread of roughly 0.0013 across all four models), supporting the conclusion that **machine learning models achieve the best predictive performance on this dataset, but by a modest margin, and traditional GLMs remain highly relevant in actuarial pricing practice because of their interpretability, statistical inference capabilities, and regulatory transparency**. Model selection should therefore be guided primarily by whether predictive accuracy or interpretability is the dominant objective for a given pricing application, rather than by predictive accuracy alone.

The identified risk factors align descriptively with rating variables used in Saudi motor insurance pricing practice as documented by the Saudi Insurance Authority (2024), supporting the potential relevance of this modelling framework beyond the French market, though this correspondence is at the level of variable categories rather than a quantitative test and should be validated with local data before being relied upon. Future research should prioritise three directions: first, extending the framework to claim severity modelling and combining frequency and severity predictions into pure premium estimates; second, applying the methodology, including the repeated-split validation approach demonstrated here, to policy-level data from additional national markets, including Saudi Arabia, to assess the generalisability of these findings; and third, incorporating SHAP-based interpretability tools (Lundberg & Lee, 2017) to help bridge the predictive advantage of the machine learning models with the transparency requirements of actuarial practice and insurance regulation.

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Appendix A: Python Code for Model Estimation

This appendix presents the Python code used to implement the models described in Section 4. All scripts use the freMTPL2freq dataset loaded from a local CSV file. The analysis was conducted using Python 3 with the pandas, statsmodels, scipy, scikit-learn, and xgboost libraries.

A.1 Distributional Assessment

```
import pandas as pd
import numpy as np

df = pd.read_csv("data/freMTPL2freq.csv")
print("Mean ClaimNb:", df["ClaimNb"].mean())
print("Variance ClaimNb:", df["ClaimNb"].var())
print("Dispersion Index:", df["ClaimNb"].var() / df["ClaimNb"].mean())
print("Proportion of zeros:", (df["ClaimNb"] == 0).mean())
```

A.2 Poisson GLM

```
import statsmodels.api as sm
import statsmodels.formula.api as smf
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error

train_df, test_df = train_test_split(df, test_size=0.2, random_state=42)
poisson_model = smf.glm(
    formula="ClaimNb ~ BonusMalus + DrivAge + VehAge + Density",
    data=train_df, family=sm.families.Poisson(),
    offset=np.log(train_df["Exposure"])).fit()
test_pred = poisson_model.predict(test_df, offset=np.log(test_df["Exposure"]))
print("MAE:", mean_absolute_error(test_df["ClaimNb"], test_pred))
print("AIC:", poisson_model.aic)
```

A.3 Negative Binomial GLM (NB2, α Estimated via Profile-Likelihood MLE)

```
from scipy.optimize import minimize_scalar
from scipy.special import gammaln

def nb2_neg_loglik(alpha, y, mu):
    r = 1.0 / alpha
    ll = (gammaln(y + r) - gammaln(r) - gammaln(y + 1)
          + r * np.log(r / (r + mu)) + y * np.log(mu / (r + mu)))
    return -np.sum(ll)

def fit_nb_fixed_alpha(train_df, alpha):
    return smf.glm(
        formula="ClaimNb ~ BonusMalus + DrivAge + VehAge + Density",
        data=train_df, family=sm.families.NegativeBinomial(alpha=alpha),
        offset=np.log(train_df["Exposure"])).fit()

# Alternating profile-likelihood estimation of alpha (mirrors R's glm.nb)
```

```

y = train_df["ClaimNb"].values
mu = poisson_model.fittedvalues.values
alpha = 1.0
for i in range(15):
    res = minimize_scalar(nb2_neg_loglik, args=(y, mu),
                        bounds=(1e-6, 20.0), method="bounded",
                        options={"xatol": 1e-10})
    alpha_new = float(res.x)
    nb_model = fit_nb_fixed_alpha(train_df, alpha=alpha_new)
    mu_new = nb_model.fittedvalues.values
    if abs(alpha_new - alpha) < 1e-6:
        alpha = alpha_new
        break
alpha, mu = alpha_new, mu_new

# AIC corrected for alpha as an estimated (not fixed) parameter
joint_loglik = -nb2_neg_loglik(alpha, y, mu)
k_params = len(nb_model.params) + 1 # +1 for alpha
aic_correct = -2 * joint_loglik + 2 * k_params

```

A.4 Random Forest

```

from sklearn.ensemble import RandomForestRegressor

features = ["Exposure", "BonusMalus", "DrivAge", "VehAge", "Density"]
rf = RandomForestRegressor(n_estimators=100, max_depth=10, random_state=42, n_jobs=-1)
rf.fit(train_df[features], train_df["ClaimNb"])
print("RF MAE:", mean_absolute_error(test_df["ClaimNb"], rf.predict(test_df[features])))

```

A.5 XGBoost

```

from xgboost import XGBRegressor

xgb = XGBRegressor(n_estimators=300, learning_rate=0.05, max_depth=4,
                  subsample=0.8, colsample_bytree=0.8, random_state=42)
xgb.fit(train_df[features], train_df["ClaimNb"])
print("XGB MAE:", mean_absolute_error(test_df["ClaimNb"], xgb.predict(test_df[features])))

```

A.6 Repeated-Split Validation

```

results = []
for seed in range(20):
    tr, te = train_test_split(df, test_size=0.2, random_state=seed)
    assert (tr["Exposure"] > 0).all() and (te["Exposure"] > 0).all()
    for name, mae in [
        ("Poisson", fit_and_eval_poisson(tr, te)),
        ("NegBinomial_NB2", fit_and_eval_nb2(tr, te)),
        ("RandomForest", fit_and_eval_rf(tr, te)),
        ("XGBoost", fit_and_eval_xgb(tr, te)),
    ]:
        results.append({"seed": seed, "model": name, "mae": mae})
# Aggregate: mean/std MAE, win count, and average rank per model (Table 8)

```

Appendix B: Repeated-Split Validation — Detailed Statistics

Table B1. Repeated-Split Validation — Full Summary Statistics (n = 20 splits)

Model	Mean MAE	Std. Dev.	Min MAE	Max MAE	95% CI (mean MAE)
Poisson GLM	0.098973	0.000396	0.098379	0.099677	[0.098788, 0.099159]
Negative Binomial GLM (α estimated)	0.099520	0.000392	0.098943	0.100206	[0.099336, 0.099703]
Random Forest	0.098202	0.000402	0.097494	0.099062	[0.098014, 0.098391]
XGBoost	0.098275	0.000388	0.097595	0.099082	[0.098093, 0.098457]

The 95% confidence intervals are constructed from the across-split sampling distribution of each model's mean test MAE. The confidence intervals for the two tree-based models do not overlap with those of either GLM, while the two tree-based models' intervals overlap substantially with each other, consistent with the two-tier pattern (tree-based models vs. GLMs) reported in Section 5.5.

Table B2. Negative Binomial α Estimates Across the 20 Repeated Splits

Split Seed	Estimated α	Converged	Test MAE
0	0.9101	Yes	0.099453
1	0.9475	Yes	0.098992
2	0.9327	Yes	0.099088
3	0.9938	Yes	0.100009
4	0.9863	Yes	0.099590
5	0.9349	Yes	0.100206
6	0.9537	Yes	0.099844
7	0.9449	Yes	0.099803
8	0.9833	Yes	0.099631
9	0.9984	Yes	0.099105
10	0.9877	Yes	0.100049
11	0.9714	Yes	0.099703
12	1.0060	Yes	0.098943
13	0.9684	Yes	0.099018
14	0.9831	Yes	0.099566
15	0.9860	Yes	0.099420
16	0.8798	Yes	0.099670
17	0.9622	Yes	0.099840
18	0.9507	Yes	0.098951
19	1.0113	Yes	0.099513

Mean α = 0.965; SD = 0.033; Min = 0.880 (seed 16); Max = 1.011 (seed 19); 20/20 splits converged.
Source: `data/repeated\_split\_results.csv`, generated by `validation\_robustness.py`.